

10) Simulate the following Disk scheduling algorithms. a) C-SCAN b) LOOK
c) C-LOOK

a) C-SCAN (Circular SCAN)

Code:

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    int n, head, diskSize;
    cout << "Enter number of requests: ";
    cin >> n;
    vector<int> req(n);
    cout << "Enter request sequence:\n";
    for(int i = 0; i < n; i++) cin >> req[i];
    cout << "Enter initial head position: ";
    cin >> head;
    cout << "Enter disk size: ";
    cin >> diskSize;
    vector<int> left, right;
    for(int i = 0; i < n; i++) {
        if(req[i] < head) left.push_back(req[i]);
        else right.push_back(req[i]);
    }
    sort(left.begin(), left.end());
    sort(right.begin(), right.end());
    int seek = 0;
    cout << "\nSeek Sequence: " << head;
```

```
// Move right
for(int x : right) {
    seek += abs(head - x);
    head = x;
    cout << " -> " << head;
}

// Go to end
seek += abs(head - (diskSize - 1));
head = diskSize - 1;

// Jump to beginning
seek += (diskSize - 1);
head = 0;
cout << " -> " << head;

// Continue right side
for(int x : left) {
    seek += abs(head - x);
    head = x;
    cout << " -> " << head;
}

cout << "\nTotal Seek Time: " << seek << endl;
}
```

Output:

Enter number of requests: 7

Enter request sequence: 82 170 43 140 24 16 190

Enter initial head position: 50

Enter disk size: 200

Seek Sequence: 50 -> 82 -> 140 -> 170 -> 190 -> 199 -> 0 -> 16 -> 24 -> 43

Total Seek Time: 391

b) LOOK

Code:

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    int n, head;
    cout << "Enter number of requests: ";
    cin >> n;
    vector<int> req(n);
    cout << "Enter request sequence:\n";
    for(int i = 0; i < n; i++) cin >> req[i];
    cout << "Enter initial head position: ";
    cin >> head;
    vector<int> left, right;
    for(int i = 0; i < n; i++) {
        if(req[i] < head) left.push_back(req[i]);
        else right.push_back(req[i]);
    }
    sort(left.begin(), left.end());
    sort(right.begin(), right.end());
    int seek = 0;
    cout << "\nSeek Sequence: " << head;
    // Move right
    for(int x : right) {
        seek += abs(head - x);
        head = x;
    }
```

```
        cout << " -> " << head;
    }
    // Move left
    for(int i = left.size()-1; i >= 0; i--) {
        seek += abs(head - left[i]);
        head = left[i];
        cout << " -> " << head;
    }
    cout << "\nTotal Seek Time: " << seek << endl;
}
```

Output:

Enter number of requests: 7

Enter request sequence: 82 170 43 140 24 16 190

Enter initial head position: 50

Seek Sequence: 50 -> 82 -> 140 -> 170 -> 190 -> 43 -> 24 -> 16

Total Seek Time: 314

c) C-LOOK**Code:**

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    int n, head;
    cout << "Enter number of requests: ";
    cin >> n;
    vector<int> req(n);
    cout << "Enter request sequence:\n";
```

```
for(int i = 0; i < n; i++) cin >> req[i];

cout << "Enter initial head position: ";

cin >> head;

vector<int> left, right;

for(int i = 0; i < n; i++) {

    if(req[i] < head) left.push_back(req[i]);

    else right.push_back(req[i]);

}

sort(left.begin(), left.end());

sort(right.begin(), right.end());

int seek = 0;

cout << "\nSeek Sequence: " << head;

// Move right

for(int x : right) {

    seek += abs(head - x);

    head = x;

    cout << " -> " << head;

}

// Jump to lowest request

if(!left.empty()) {

    seek += abs(head - left[0]);

    head = left[0];

    cout << " -> " << head;

}

// Continue right

for(int i = 1; i < left.size(); i++) {

    seek += abs(head - left[i]);

    head = left[i];

}
```

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Name of the laboratory:

Name of the Experiment: _____

Experiment No: _____ **Date:** _____

```
cout << " -> " << head;
```

```
}
```

```
cout << "\nTotal Seek Time: " << seek << endl;
```

```
}
```

Output:

Enter number of requests: 7

Enter request sequence:

82 170 43 140 24 16 190

Enter initial head position: 50

Seek Sequence: 50 -> 82 -> 140 -> 170 -> 190 -> 16 -> 24 -> 43

Total Seek Time: 341